THE RIGHT ADJOINT OF PUSHFORWARD ALONG A PSEUDO-COHERENT CLOSED IMMERSION IS THE EXACT-SUPPORT SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A9X (sources/stacks-duality/pinned-original-source.tex line 2467).

Allowed standard prerequisites (editorial).

- The exact support sheaf is right adjoint to pushforward. Same chapter, Lemma labelled lemma-sheaf-with-exact-support-adjoint, cited at proof lines 2481-2485. Shows RSheafHom(O_Z, -) is a right adjoint of Ri_*.
- Quasi-coherence of the exact support sheaf. Same chapter, Lemma labelled lemma-sheaf-with-exact-support-quasi-coherent, cited at proof lines 2485-2489. Keeps the comparison inside D_QCoh.
- Bounded belowness of the twisted inverse image. Same chapter, Lemma labelled lemma-twisted-inverse-image-bounded-below, cited at proof lines 2485-2489. Keeps the comparison inside D^+.

Source reference. Original lemma 0684: sources/stacks-chow/tags.txt, source lines 7993–7993. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2481 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A76: sources/stacks-duality/pinned-original-source.tex, source lines 2325–2325. Same-chapter pointer into the author environment lemma at source line 2325; invoked at proof line 2482 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A78: sources/stacks-duality/pinned-original-source.tex, source lines 2373–2373. Same-chapter pointer into the author environment lemma at source line 2373; invoked at proof line 2485 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9I: sources/stacks-duality/pinned-original-source.tex, source lines 726–726. Same-chapter pointer into the author environment lemma at source line 726; invoked at proof line 2486 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $i : Z \to X$ be a pseudo-coherent closed immersion (if X is Noetherian, then any closed immersion is pseudo-coherent). Let $a : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Z)$ be the right adjoint to Ri_*. Then there is a functorial isomorphism*

$$a(K) = R\mathcal{H}om(\mathcal{O}_Z, K)$$

for $K \in D^+_{QCoh}(\mathcal{O}_X)$.

Proof. (The parenthetical statement follows from More on Morphisms, Lemma 0684.) By Lemma 0A76 the functor $R\mathcal{H}om(\mathcal{O}_Z, -)$ is a right adjoint to $Ri_* : D(\mathcal{O}_Z) \to D(\mathcal{O}_X)$. Moreover, by Lemma 0A78 and Lemma 0A9I both $R\mathcal{H}om(\mathcal{O}_Z, -)$ and a map $D^+_{QCoh}(\mathcal{O}_X)$ into $D^+_{QCoh}(\mathcal{O}_Z)$. Hence we obtain the isomorphism by uniqueness of adjoint functors. □